

Human pathogens and tetracycline-resistant bacteria in bioaerosols of swine confinement buildings and in nasal flora of hog producers.

Swine confinement buildings in eastern Canada are enclosed and equipped with modern production systems to manage waste. Bioaerosols of these swine confinement buildings could be contaminated by human pathogens and antimicrobial resistant bacteria which could colonize exposed workers. We therefore wanted to analyze bioaerosols of swine confinement buildings and nasal flora of Canadian hog producers to evaluate possible colonization with human pathogens and tetracycline-resistant bacteria. Culturable and non-culturable human pathogens and tet genes were investigated in the bioaerosols of 18 barns. The nasal passages of 35 hog producers were sampled and total DNA was extracted from the calcium-alginate swabs to detect, by PCR, *Campylobacter*, *C. perfringens*, *Enterococcus*, *E. coli*, *Y. enterocolitica*, *tetA/tetC*, *tetG* and ribosomal protection protein genes. Airborne culturable *C. perfringens*, *Enterococcus*, *E. coli*, and *Y. enterocolitica* were present in the bioaerosols of 16, 17, 11 and 6 of the 18 facilities. Aerosolized total (culturable/non culturable) *Campylobacter*, *C. perfringens*, *Enterococcus*, *E. coli* and *Y. enterocolitica* were detected in 10, 6, 15, 18 and 2 barns, respectively. Tet genes were found in isolates of culturable human pathogens. *TetA/tetC*, *tetG* and ribosomal protection protein genes were detected in the bioaerosols of all 18 studied buildings. *Campylobacter*, *C. perfringens*, *Enterococcus*, *E. coli*, and *Y. enterocolitica* were found respectively in 4, 9, 17, 14 and one nasal flora of workers. One and 10 workers were positive for *tetA/tetC* and *tetG* genes, respectively. In swine confinement buildings, hog producers are exposed to aerosolized human pathogens and tetracycline-resistant bacteria that can contaminate the nasal flora. Copyright © 2010 Elsevier GmbH. All rights reserved.